

Quasi-Likelihood and Overdispersion

A generalized linear model asks for a great deal and gives a great deal back. Naming the family fixes the whole distribution of the response, and with it the exact relation between its mean and its variance — the part of the bargain the data most often refuse. Counts of accidents, litters of affected pups, votes in households, durations of strikes are nearly always more variable than the Poisson or the binomial allows, and no choice of link repairs it. This chapter takes the least drastic way out. The likelihood equations of [Chapter 34](#) use the density only through the mean and the variance function, so one may write down those two and estimate from the equations alone. The estimate is unchanged, its consistency survives a wrong variance function, and its standard errors can be read off the data rather than the model. What is given up is the likelihood, and with it the likelihood ratio test, the Akaike criterion, and every statement about a probability.

What you need. [Chapter 34](#) throughout: the model (Definition 34.2.1), the variance function (Definition 34.1.2), the score and the information (Theorem 34.3.1), iteratively reweighted least squares (Theorem 34.4.1), the deviance (Definition 34.5.1, Theorem 34.5.5) and the asymptotics (Theorem 34.5.1, Proposition 34.5.2). [Section 21.4](#) for the sandwich estimators (Definition 21.4.1, Theorem 21.4.2) and [Section 21.3](#) for weighted least squares (Theorem 21.3.1). [Chapter 37](#) for the negative binomial models (Definition 37.4.1, Proposition 37.4.2) and the score test for overdispersion (Theorem 37.4.3). The optimality proof of [Section 38.2](#) uses the projection inequality of [Chapter 6](#); [Chapter 20](#), [Chapter 23](#), [Chapter 29](#) and [Chapter 33](#) are used in single sections.

38.-1.1 Roadmap

- [Variance functions](#)
- [Quasi-likelihood estimating equations](#)
- [Sandwich covariance estimation](#)
- [Overdispersion](#)
- [Summary and notes](#)